

IM-DD Net Bit Rate Beyond 800 Gb/s per Lane with Single-Photodiode Reception of 440-GBaud PAM

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Abstract We demonstrate single-lane intensity modulation and direct detection with a net bit rate of 827 Gb/s using a single modulator and a single photodiode. A new 3-band digital-band-interleaved receiver covering 221-GHz bandwidth is built on a 512-GSa/s real-time oscilloscope. ©2026 The Author(s)

Introduction

Intensity modulation with direct detection (IM-DD) is the mainstream technology for the optical interconnects inside datacenters, thanks to its low cost, power consumption, and latency. With 200G-per-lane IM-DD optics now commercial and 400G per lane being heavily discussed for next-generation 1.6T and 3.2T transceivers [1], it remains an open question whether the next capacity doubling to 800G per lane can still be served by a single wavelength, a single modulator and a single photodiode.

Carrying a net 800 Gb/s on one IM-DD lane with pulse amplitude modulation (PAM) formats of ~2 information bits/symbol requires a symbol rate in the 400-GBaud class, which translates into electro-optic (E/O) and opto-electronic (O/E) bandwidths beyond 200 GHz. On the E/O side, multiple modulator platforms are being actively developed toward this class, like thin-film lithium niobate [1,2], thin-film lithium tantalate [3], barium titanate [4], and plasmonics [5]. The O/E side [6-8] has received far less attention, yet it faces a critical trade-off between bandwidth and responsivity: a fast photodiode demands a thin absorber and a small junction, which sacrifice absorption and optical coupling. As a result, 200-GHz-class photodiodes have long delivered responsivities of only a few tenths of an A/W [6,7]. This trade-off

was broken recently by a waveguide-integrated modified uni-travelling-carrier (MUTC) photodiode that set a bandwidth–efficiency product benchmark of 133 GHz, combining a 206-GHz bandwidth with a 0.81-A/W external responsivity [8], which is the O/E enabler of the present work.

Beyond the optical devices, the tighter limitation is electronic: commercial D/A and A/D converters (DACs and ADCs) offer barely half of the analog bandwidth that a 400-GBaud-class lane demands. Fig. 1 summarizes the reported high-speed IM-DD demonstrations in terms of net bit rate (NBR) versus Nyquist symbol rate [1-5,9-13]. On the transmitter side, the leading experiments overcame the DAC limitation by electrical multiplexing [9,11-13], and digital-band-interleaved (DBI) DACs have pushed all-electronic signal generation up to 440 GBaud [15,16]. The receiver has proven equally hard to extend: electrical bandwidth demultiplexing has been demonstrated for 100-GBaud-class reception [14], and combining it with transmitter-side multiplexing using 150-GHz mixers yielded the record NBR of 660 Gb/s at 248 GBaud [12], while the 314- and 440-GBaud signals could so far only be received through coherent-detection-based optical spectral stitching [15,16]. As Fig. 1 shows, IM-DD has thus plateaued at the 250-GBaud class limited by the receiver bandwidth. In this paper, we break this ceiling with a single-carrier 440-GBaud IM-DD system whose electrical spectrum is detected by a single MUTC photodiode and digitized by a novel three-band DBI-ADC receiver with 221-GHz analog bandwidth. Using probabilistically shaped (PS) PAM, the link achieves an NBR of 826.6 Gb/s per lane, which is beyond the net-800-Gb/s mark and ~25% above the highest previously published IM-DD NBR [12].

Receiver Structure

The photodiode in this work follows the InP-based, waveguide-integrated MUTC design [8]. Its bandwidth is set jointly by the RC time constant and the carrier transit time, whereas its responsivity is set by how much light is absorbed by the active layer, and the design addresses

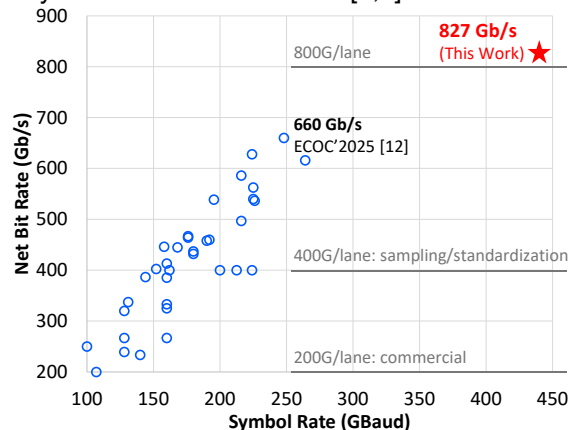


Fig. 1: IM-DD net bit rate (NBR) versus Nyquist symbol rate of reported single-lane experiments with NBR above 200 Gb/s [1-5,9-13]. Only selected points are cited for space. Horizontal lines mark 200G, 400G, and 800G-per-lane generations.

them separately. A two-step spot-size converter couples the light efficiently into an active area of only $\sim 30 \mu\text{m}^2$, so that a compact, low-capacitance junction retains high external responsivity; a heavily doped cliff layer in the InP drift layer shapes the electric field to sustain electron velocity overshoot and counteract space-charge effects at photocurrents up to 10 mA; and benzocyclobutene beneath the coplanar-waveguide electrodes minimizes the parasitic capacitance. The device used in our experiment (Fig. 2(i)) exhibits a 3-dB bandwidth of about 190 GHz. It is contacted by a ground-signal-ground probe and reverse-biased at -1.2 V through a bias-tee, and the light is coupled to its spot-size converter by a lensed fiber with a mode-field diameter of $2 \mu\text{m}$. Fig. 2(ii) shows the measured photocurrent against the continuous-wave (CW) optical power at 1550 nm; the slope of the linear fit gives an external responsivity of 0.46 A/W , which includes the coupling loss of the lensed fiber, and the response stays linear up to 7 mA.

To digitize the 440-Gbaud photodiode output spectrum, we develop an three-band DBI-ADC, the mirror image of the DBI-DAC transmitter [16]. The DBI-ADC builds on a new-generation Keysight real-time oscilloscope (XR9), which employs frequency-domain interleaving internally to reach about 180 GHz of analog bandwidth at a 512-GSa/s sampling rate per channel. The externally demultiplexed DBI-ADC both exploits the full XR9 bandwidth and extends the overall analog bandwidth to 221 GHz. The photodiode output is first split by a diplexer probe into band 1 (DC–130 GHz) and a high-frequency path (130–221 GHz); the latter is further divided by a low-pass (LPF) and a high-pass filter (HPF) into band 2 (130–180 GHz) and band 3 (175–221 GHz). Bands 1 and 2 lie within the oscilloscope's bandwidth and are detected directly at their absolute RF positions; the 130-GHz boundary between

the two bands is set by the crossover of the diplexer probe. Band 3 is down-converted externally by a 222-GHz local oscillator (LO). Mixing with an LO above the band maps 175–221 GHz onto the 1–47 GHz lower sideband, with the reversed spectrum flipped back digitally. Although the band-3 HPF passes frequencies from 175 GHz, it is band-limited by digital signal processing (DSP) from 180 GHz, demonstrating that spectral overlap between bands is not a prerequisite for the DBI-ADC. The three channels share the oscilloscope's sampling clock. In the digital domain, each slice is band-limited and passed through a frontend correction that equalizes the frequency response of its RF path, normalizes its power, and matches the inter-band delays; the three slices are then frequency-shifted back to their original spectral positions and jointly equalized to reconstruct the full DC–221-GHz waveform.

Experimental Setup

Fig. 2 shows the experimental setup. The transmitter is the 221-GHz DBI-DAC reported in [16]: four spectral bands covering DC–90, 90–130, 130–175, and 175–221 GHz are generated by a 4-channel 256-GSa/s arbitrary waveform generator (Keysight M8199B), up-converted by local oscillators at 87, 128, and 222 GHz synthesized from a common clock, and combined into a continuous DC–221-GHz spectrum by passive combiners and a diplexer with a crossover frequency of about 130 GHz. The frequency responses and path delays of all RF components are pre-compensated digitally per band [16] (transmitter DSP in Fig. 2(v)). A CW laser at 1550 nm, amplified by an EDFA to 23 dBm, is modulated by a thin-film LiNbO₃ Mach-Zehnder modulator (MZM) with a 6-dB bandwidth of $>110 \text{ GHz}$ and an insertion loss of 8 dB, biased off the null point for intensity modulation; the DBI-DAC output drives the MZM through an RF probe integrated with the diplexer.

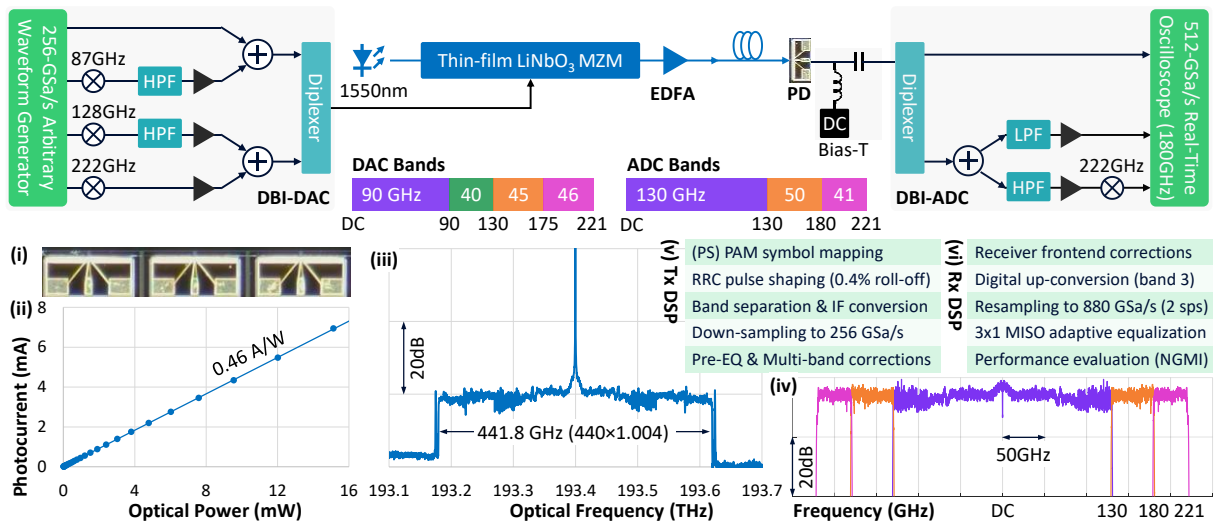


Fig. 2: Experimental setup with the 221-GHz DBI-DAC transmitter and three-band DBI-ADC receiver. Insets: (i) photodiode micrograph (showing three devices); (ii) measured photocurrent vs. optical power (0.46 A/W); (iii) optical spectrum at the MZM output; (iv) recombined DC–221-GHz spectrum after DBI-ADC processing; (v) transmitter (Tx) and (vi) receiver (Rx) DSP.

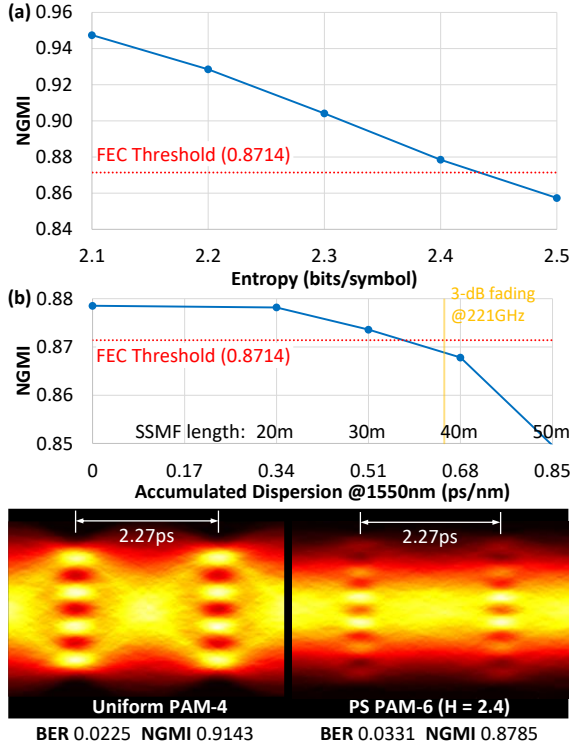


Fig. 3: (a) NGMI vs. entropy H (back-to-back); (b) NGMI vs. accumulated CD at 1550 nm (PS PAM-6, $H = 2.4$); the vertical line marks the 0.65-ps/nm CD that causes 3-dB fading at the 221-GHz band edge; bottom: recovered eye diagrams.

Fig. 2(iii) shows the optical spectrum at the modulator output, which is almost flat across 440-GHz optical bandwidth without any optical equalization. The signal is transmitted either back-to-back or over standard single-mode fiber (SSMF), and then detected by the MUTC photodiode described above. The output spectrum is captured by the three-band DBI-ADC, whose recombined electrical spectrum after the DBI-ADC processing is shown in Fig. 2(iv).

In the receiver DSP (Fig. 2(vi)), the three slices are frequency-converted to their assigned bands and resampled to 2 samples/symbol (sps). Following frame synchronization, the three bands are fed to a frequency-domain 3×1 multi-input, single-output (MISO) equalizer with 12.5-ns time duration. The adaptive MISO filter equalizes the channel while simultaneously correcting the residual amplitude and phase mismatch among the three ADC bands. The bands could equally be summed before entering a conventional SISO equalizer if dedicated inter-band calibration is performed. Note that no nonlinear equalization is used in any result reported below.

Results

We generate 440-GBaud PAM signals with root-raised-cosine filtering (0.4% roll-off), occupying an electrical bandwidth of $440 \times 1.004/2 = 220.9$ GHz. Three format families are measured: uniform PAM-4; PS PAM-6, the inner six levels of the PAM-8 alphabet under Maxwell-Boltzmann

shaping with entropies (H) from 2.1 to 2.4; and PS PAM-8 with $H = 2.5$. All results are taken at a photocurrent of 5 mA. We choose a rate-0.8262 forward error correction (FEC) scheme that concatenates a soft-decision spatially-coupled low-density parity-check (SC-LDPC) inner code (rate 5/6) and a hard-decision BCH outer code (rate 0.9915), with a normalized generalized mutual information (NGMI) threshold of 0.8714 [17]. The NBR is computed as

$$NBR = B \cdot (H - (1 - c) \cdot \log_2 M)$$

where B is the symbol rate, c is the FEC code rate, and M (4 or 8) is the modulation order. Fig. 3(a) shows the NGMI versus entropy in back-to-back configuration. The measured SNR is about 12.4 dB for all the formats, confirming the effectiveness of PS formats in this IM-DD system [18]. The highest entropy whose NGMI exceeds the FEC threshold is 2.4 bits/symbol yielding an NBR of 826.6 Gb/s. The eye diagrams show the recovered PAM-4 and PS-PAM-6 ($H = 2.4$) signals.

For double-sideband IM signals, the square-law detection turns the chromatic dispersion (CD) induced phase rotation of the two sidebands into frequency selective power fading, scaling with the square of the frequency. At the 220.9-GHz band edge, 3-dB fading occurs at 0.65 ps/nm and the first null at 1.28 ps/nm, *i.e.*, after 38 m and 76 m of SSMF with $D \approx 17$ ps/nm/km at C-band. Fig. 3(b) is consistent with this fading picture: the 826.6-Gb/s NBR is retained up to 30 m, and the SNR penalty grows to about 0.7 dB at 50 m. Taking the 3-dB band-edge fading (0.65-ps/nm CD) as a benchmark, a 400-GBaud-class IM-DD can only support C-band SSMF over tens of meters. The same 3-dB fading criterion allows ≈ 0.9 ps/nm in the O-band (as the fading scales with λ^2), *i.e.*, ≈ 900 m at $|D| = 1$ ps/nm/km and several kilometers close to the zero-dispersion wavelength. Alternatively, it corresponds to ≈ 200 m of hollow-core fiber with $D \approx 3$ ps/nm/km at 1550 nm [19]. A single-lane 800G IM-DD is thus CD-compatible with the sub-km scale-up domain of AI clusters, with longer reach calling for operation near zero-dispersion wavelengths or CD-tolerant signaling.

Conclusions

We have demonstrated a single-lane 440-GBaud IM-DD transmission at an NBR of 827 Gb/s, received by a single photodiode with all-electronic demultiplexing rather than the optical spectral stitching that previous demonstrations have relied on at this symbol rate. This is achieved with the mainstream architecture of one polarization, one modulator and one photodiode, showing that the per-lane scaling that carried IM-DD from 25G to 200G does not end at 400G. Such sustained rate doubling allows IM-DD to preserve its cost, power and footprint advantage as datacenter interfaces keep growing.

References

- [1] C. St-Arnault, S. Bernal, D. Kita, R. Dickson, M. Y. Abdelaziz, A. Nikic, B. Qiu, B. Krueger, F. Pittalà, C. Reimer, B. Beggs, N. Ben-Hamida, and D. V. Plant, "Net 3.2 Tbps 225 Gbaud PAM4 O-Band IM/DD 2 km Transmission Using FR8 and DR8 with a CMOS 3 nm SerDes and TFLN Modulators," in Optical Fiber Communication Conference (OFC) 2025, postdeadline paper Th4B.1, DOI: 10.1364/OFC.2025.Th4B.1.
- [2] E. Berikaa, M. S. Alam, W. Li, S. Bernal, B. Krueger, F. Pittalà, and D. V. Plant, "TFLN MZMs and Next-Gen DACs: Enabling Beyond 400 Gbps IMDD O-Band and C-Band Transmission," IEEE Photonics Technology Letters, vol. 35, no. 15, pp. 850-853, 2023, DOI: 10.1109/LPT.2023.3285881.
- [3] M. Xu, Y. Lan, D. Che, J. Shi, and D. Liang, "C-Band 110-GHz-Bandwidth Thin-Film Lithium Tantalate Modulator Enabling 768 (536) Gbit/s Line (Net) Data Rates," in Optical Fiber Communication Conference (OFC) 2026, postdeadline paper Th4A.2, DOI: 10.1364/OFC.2026.Th4A.2.
- [4] B. Qiu, C. St-Arnault, J. Cauchon, C. Minkenberg, T. Kornher, C. Laperle, W. Diels, S. Abel, N. Ben-Hamida, B. Beggs, F. Pelletier, L. Czornomaz, X. Chen, F. Eltes, and D. V. Plant, "Barium Titanate Enabling Net 1.6T (4x448 Gbps PAM4) on a Silicon Photonics Platform," in Optical Fiber Communication Conference (OFC) 2026, postdeadline paper Th4B.3, DOI: 10.1364/OFC.2026.Th4B.3.
- [5] L. Kulmer, S. Oettinghaus, S. Hess, M. Destraz, A. Dochhan, S. Pachnicke, B. Baeuerle, C. Hoessbacher, W. Heni, T. Blatter, and J. Leuthold, "O-Band Plasmonic MZM Enabling Single Carrier net 400 Gbit/s IM/DD Over 1 km Fiber," in European Conference on Optical Communication (ECOC) 2025, DOI: 10.1109/ECOC66593.2025.11263107.
- [6] H. Ito, T. Furuta, S. Kodama, and T. Ishibashi, "InP/InGaAs uni-travelling-carrier photodiode with 310 GHz bandwidth," Electronics Letters, vol. 36, no. 21, pp. 1809-1810, 2000, DOI: 10.1049/el:20001274.
- [7] S. Lischke, A. Peczek, J. S. Morgan, K. Sun, D. Steckler, Y. Yamamoto, F. Korndörfer, C. Mai, S. Marschmeyer, M. Fraschke, A. Krüger, A. Beling, and L. Zimmermann, "Ultra-fast germanium photodiode with 3-dB bandwidth of 265 GHz," Nature Photonics, vol. 15, pp. 925-931, 2021, DOI: 10.1038/s41566-021-00893-w.
- [8] L. Li, T. Long, X. Yang, Z. Zhang, L. Wang, J. Wang, M. Wang, J. Lu, J. Yu, and B. Chen, "Modified uni-travelling-carrier photodiodes with 206 GHz bandwidth and 0.81 A W⁻¹ external responsivity," Nature Photonics, vol. 19, pp. 1301-1308, 2025, DOI: 10.1038/s41566-025-01784-0.
- [9] H. Yamazaki, M. Nakamura, T. Kobayashi, M. Nagatani, H. Wakita, Y. Ogiso, H. Nosaka, T. Hashimoto, and Y. Miyamoto, "Net-400-Gbps PS-PAM transmission using integrated AMUX-MZM," Optics Express, vol. 27, no. 18, pp. 25544-25550, 2019, DOI: 10.1364/OE.27.025544.
- [10] X. Chen, J. Cho, G. Raybon, D. Che, K. W. Kim, E. Burrows, P. Kharel, C. Reimer, K. Luke, L. He, and M. Zhang, "Single-Wavelength and Single-Photodiode 700 Gb/s Entropy-Loaded PS-256-QAM and 200-GBaud PS-PAM-16 Transmission over 10-km SMF," in European Conference on Optical Communication (ECOC) 2020, postdeadline paper Th3A-2, DOI: 10.1109/ECOC48923.2020.9333201.
- [11] M. Nakamura, T. Jyo, M. Nagatani, H. Wakita, M. Mutoh, Y. Shiratori, H. Taniguchi, A. Masuda, S. Yamamoto, F. Hamaoka, E. Yamazaki, H. Takahashi, T. Kobayashi, and Y. Miyamoto, "Net 582-Gb/s C-band and 4x526-Gb/s O-band IMDD Transmission Using Ultra-broadband InP-DHBT-based Electrical Mixer," in European Conference on Optical Communication (ECOC) 2024, postdeadline paper Th3B.3.
- [12] M. Nakamura, M. Nagatani, T. Jyo, H. Wakita, M. Mutoh, Y. Shiratori, F. Hamaoka, H. Taniguchi, S. Yamamoto, E. Yamazaki, T. Kobayashi, H. Takahashi, and Y. Miyamoto, "651-Gb/s Net Bitrate IMDD Transmission Using Electrical Bandwidth Multiplexing and Demultiplexing Techniques Based on Ultra-Broadband InP-DHBT Mixers," in European Conference on Optical Communication (ECOC) 2025, DOI: 10.1109/ECOC66593.2025.11262960.
- [13] Q. Hu, M. Grözing, T. Tannert, R. Borkowski, F. Buchali, P. Iannone, G. Raybon, X. Chen, G. Rademacher, and R. Ryf, "High-Performance SiGe Analog Multiplexer Enabling 176 GBd PAM-8 Transmission," Journal of Lightwave Technology, vol. 43, no. 4, pp. 1881-1892, 2025, DOI: 10.1109/JLT.2025.3527328.
- [14] F. Buchali, K. Schuh, S. T. Le, X.-Q. Du, M. Grözing, and M. Berroth, "A SiGe HBT BiCMOS 1-to-4 ADC Frontend Supporting 100 GBaud PAM4 Reception at 14 GHz Digitizer Bandwidth," in Optical Fiber Communication Conference (OFC) 2019, postdeadline paper Th4A.7, DOI: 10.1364/OFC.2019.Th4A.7.
- [15] D. Che, C. Deakin, and X. Chen, "314-GBaud Single-Wavelength Signaling Generated All-Electronically by a 158-GHz Digital-Band-Interleaved DAC," in European Conference on Optical Communication (ECOC) 2024, postdeadline paper Th3B.4.
- [16] D. Che, C. Deakin, X. Chen, and G. Raybon, "440-GBaud All-Electronic Signaling Enabling Single-Wavelength Net Rate Over 1 Tb/s Per Modulation Dimension," Journal of Lightwave Technology, vol. 44, no. 3, pp. 1178-1193, 2026, DOI: 10.1109/JLT.2025.3632244.
- [17] J. M. Gené, X. Chen, J. Cho, S. Chandrasekhar, and P. Winzer, "Experimental Demonstration of Widely Tunable Rate/Reach Adaptation From 80 km to 12,000 km Using Probabilistic Constellation Shaping," in Optical Fiber Communication Conference (OFC) 2020, paper M3G.3, DOI: 10.1364/OFC.2020.M3G.3.
- [18] D. Che, J. Cho, and X. Chen, "Does Probabilistic Constellation Shaping Benefit IM-DD Systems Without Optical Amplifiers?," Journal of Lightwave Technology, vol. 39, no. 15, pp. 4997-5007, 2021, DOI: 10.1109/JLT.2021.3083530.
- [19] M. Petrovich, E. Numkam Fokoua, Y. Chen, H. Sakr, A. I. Adamu, R. Hassan, D. Wu, R. Fatobene Ando, A. Papadimopoulos, S. R. Sandoghchi, G. Jasion, and F. Poletti, "Broadband optical fibre with an attenuation lower than 0.1 decibel per kilometre," Nature Photonics, vol. 19, pp. 1203-1208, 2025, DOI: 10.1038/s41566-025-01747-5.